REFLEXIVITY IS LOCAL ON THE PRIMES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Localisation of Hom out of a finitely presented module. Algebra, Lemma \ref{algebra-lemma-hom-from-finitely-presented}, cited at proof lines 5249-5251. Identifies the localised double dual map with the map for the localised module.
- A module map is an isomorphism if and only if it is so after localising at every prime. Algebra, Lemma \ref{algebra-lemma-characterize-zero-local}, cited at proof lines 5252-5253. Gives the local-to-global step.

Source reference. Original lemma 0AV1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13669–13669. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0583: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6766–6766. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian domain. Let M be a finite R-module. The following are equivalent:*

(1) *M is reflexive,*

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(2) $M_{\mathfrak{p}}$ *is a reflexive* $R_{\mathfrak{p}}$*-module for all primes* $\mathfrak{p} \subset R$*, and*
(3) $M_{\mathfrak{m}}$ *is a reflexive* $R_{\mathfrak{m}}$*-module for all maximal ideals* $\mathfrak{m}$ *of* R*.*

Proof. The localization of $j : M \to \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)$ at a prime $\mathfrak{p}$ is the corresponding map for the module $M_{\mathfrak{p}}$ over the Noetherian local domain $R_{\mathfrak{p}}$. See Algebra, Lemma 0583. Thus the lemma holds by Algebra, Lemma 00HN. □